



clinical trials tests performed on human subjects

research subjects to decide if they want to participate. It is important both to consider the safety of the experimental treatment and to understand the risks and benefits to the subjects. In utero gene therapy has the added complexity of posing risks not only to the fetus, but also to the pregnant woman. Further, voluntary consent is imperative. Gene therapy may be the only possible treatment, or the treatment of last resort, for some individuals. In such cases, it becomes questionable whether the patient can truly be said to make a voluntary decision to participate in the trial.

Gene therapy **clinical trials** came under scrutiny in September 1999, after the highly publicized death of a gene therapy clinical trial participant several days after he received the experimental treatment. This case raised concerns about the overall protection of human subjects in clinical testing, and specifically about the reliability of the informed consent process. In this case, it was alleged that information about potential risks to the patient was not fully disclosed to the patient and his family. It was further alleged that full information regarding adverse events (serious side effects or deaths) that occurred in animals receiving experimental treatment had not been adequately disclosed. Adverse events should be disclosed in a timely manner not only to the participants in these trials, but also to the regulatory bodies overseeing gene therapy clinical trials. Furthermore, participants had not been told of a conflict of interest posed by a financial relationship between the university researchers and the company supporting the research. Obviously, any conflicts of interests could interfere with the objectivity of researchers in evaluating the effectiveness of the clinical trials and should be disclosed during the informed consent process.

Appropriate Uses of Gene Therapy

How do researchers determine which disorders or traits warrant gene therapy? Unfortunately, the distinction between gene therapy for disease genes and gene therapy to enhance desired traits, such as height or eye color, is not clear-cut. No one would argue that diseases that cause suffering, disability, and, potentially, death are good candidates for gene therapy. However, there is a fine line between what is considered a “disease” (such as the dwarfism disorder achondroplasia) and what is considered a “trait” in an otherwise healthy individual (such as short stature). Even though gene therapy for the correction of potentially socially unacceptable traits, or the enhancement of desirable ones, may improve the quality of life for an individual, some ethicists fear gene therapy for trait enhancement could negatively impact what society considers “normal” and thus promote increased discrimination toward those with the “undesirable” traits. As the function of many genes continue to be discovered, it may become increasingly difficult to define which gene traits are considered to be diseases versus those that should be classified as physical, mental, or psychological traits.

ethicists a person who writes and speaks about ethical issues

germ line cells giving rise to sperm or eggs

To date, acceptable gene therapy clinical trials involve somatic cell therapies using genes that cause diseases. However, many **ethicists** worry that, as the feasibility of **germ line** gene therapy improves and more genes causing different traits are discovered, there could be a “slippery slope” effect in regard to which genes are used in future gene therapy experiments. Specifically, it is feared that the acceptance of germ line gene therapy could lead to the acceptance of gene therapy for genetic enhancement. Public debate

about the issues revolving around germ line gene therapy and gene therapy for trait enhancement must continue as science advances to fully appreciate the appropriateness of these newer therapies and to lead to ethical guidelines for advances in gene therapy research.

Promise of Gene Therapy

There was extensive media coverage and public excitement about the promise of gene therapy as the first clinical trials commenced in 1990. In fact, many individuals and families with genetic disorders expected an imminent cure for their diseases. Unfortunately, there are as yet few successes to report, even though hundreds of somatic cell gene therapy clinical trials for many different diseases have been attempted. The early hype of gene therapy might have been avoided with more open and honest communication about gene therapy and its expectations between the researchers, physicians, patients, and the public. The death of the participant in 1999 and deficiencies in the protocol (study design) used for that trial underscore the need for continued public discourse on gene therapy.

The promise of gene therapy has not diminished, even though its full therapeutic potential is not yet known. Scientists, physicians, patients, and families continue to look forward to many future successes for gene therapy. With the completion of the Human Genome Project and the accelerated discovery of human disease genes, the potential number of diseases for which gene therapy could be beneficial continues to increase. With further research into the technical aspects of gene therapy and continued public debate about the ethical issues involved in such treatments, it is hoped gene therapies will become standard, effective treatments in the next few decades. **SEE ALSO** CYSTIC FIBROSIS; EUGENICS; GENE THERAPY; GROWTH DISORDERS; MUSCULAR DYSTROPHY; PRENATAL DIAGNOSIS; SEVERE COMBINED IMMUNE DEFICIENCY.

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 Smaglik, Paul. "Congress Gets Tough with Gene Therapy." *Nature* 403 (2000): 583–584.
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Internet Resources

- American Society of Human Gene Therapy. <<http://www.asgt.org>>.
 "Gene Therapy." Human Genome Program of the U.S. Department of Energy. <<http://www.ornl.gov/hgmis/medicine/genetherapy.html>>.

Genetic Code

The sequence of **nucleotides** in DNA determines the sequence of amino acids found in all proteins. Since there are only four nucleotide "letters" in the DNA alphabet (A, C, G, T, which stand for adenine, cytosine, guanine, and thymine), but there are 20 different amino acids in the protein alphabet, it is clear that more than one nucleotide must be used to specify an amino acid. Even two nucleotides read at a time would not give sufficient combinations ($4 \times 4 = 16$) to encode all 20 amino acids plus start and stop signals. Therefore it would require a minimum of three DNA nucleotides

nucleotides the building blocks of RNA or DNA



Figure 1. A messenger RNA is translated in triplets, beginning with the first AUG encountered by the ribosome. Translation stops at a stop codon, one of which is UAA.

mRNA sequence:	A	U	G	C	U	U	A	G	U	C	. . .	U	A	A
	[]			[]			[]			[]			[]	
Protein sequence:	Met			Pro			Leu			Val			Stop	
	(Start)													

to “spell out” one amino acid, and indeed this is the number that is actually used. RNA also uses a four letter alphabet when it reads and transcribes DNA instructions during protein synthesis, but its set of nucleotides is somewhat different, substituting U (uracil) for T (thymine).

Any single set of three nucleotides is called a **codon**, and the set of all possible three-nucleotide combinations is called “the genetic code” or “triplet code.” There are sixty-four different combinations or codons ($4 \times 4 \times 4 = 64$). We now know that three codons (UAA, UAG, and UGA) specify a “stop” signal, indicating the termination of the **polypeptide** chain being synthesized on the ribosome. Each of the remaining sixty-one codons encodes an amino acid. The “start” signal is the codon AUG, which also encodes the amino acid methionine. The codons are read from the messenger RNA molecule during protein synthesis, and, consequently, they are given in RNA bases rather than in the original DNA sequence. The reading of the codons is shown in Figure 1.

Translation

The gene is represented by the sequences of bases in the DNA molecule, which can, in a sense, be thought of as a “storage molecule” for genetic information. DNA is extremely stable, a property critical to the maintenance of the integrity of the gene. This stability is evidenced by the fact that DNA has been extracted from Egyptian mummies and extinct animals such as the woolly mammoth. It can be extracted from dried blood or from a single hair at a crime scene.

Each cell contains a complete set of genes, but only certain of these genes are active or “expressed” at any one time. When a gene is active, a “disposable” copy is transcribed from the gene into codons contained in a messenger RNA (mRNA) molecule. Unlike the DNA molecule, the mRNA molecule is relatively unstable and short-lived. This is so that when a gene is turned off, the mRNA does not remain in the cell forever, running off more proteins on the ribosomes that are no longer needed by the cell.

Another RNA molecule, called transfer RNA (tRNA), contains a specific region called the anticodon. The tRNA anticodon can base pair with the codon region of the mRNA during protein synthesis, using the base pairing rules of A-U, U-A, C-G, and G-C. Each tRNA carries a specific amino acid. Thus the tRNA carrying methionine has a UAC anticodon that pairs with the AUG codon of the mRNA bound to the ribosome. Similarly the tRNA for proline has a GGA anticodon.

In examining the table of codons (Table 1) you will see that there is more than one codon for each amino acid, except for methionine (AUG) and tryptophan (UGG). Different codons that code for the same amino acid are said to be “synonyms,” and the code is said to be “degenerate” in the sense that there is not a single, unique codon for each of the twenty amino acids.

codon a sequence of three mRNA nucleotides coding for one amino acid

polypeptide chain of amino acids

		Second base			
		U	C	A	G
First base	U	UUU } Phe	UCU } Ser	UAU } Tyr	UGU } Cys
		UUC }	UCC }	UAC }	UGC }
		UUA } Leu	UCA }	UAA } STOP	UGA } STOP
		UUG }	UCG }	UAG }	UGG } Trp
	C	CUU } Leu	CCU } Pro	CAU } His	CGU } Arg
		CUC }	CCC }	CAC }	CGC }
		CUA }	CCA }	CAA } Gln	CGA }
		CUG }	CCG }	CAG }	CGG }
	A	AUU } Ile	ACU } Thr	AAU } Asn	AGU } Ser
		AUC }	ACC }	AAC }	AGC }
		AUA } Met	ACA }	AAA } Lys	AGA } Arg
		AUG }	ACG }	AAG }	AGG }
	G	GUU } Val	GCU } Ala	GAU } Asp	GGU } Gly
		GUC }	GCC }	GAC }	GGC }
		GUA }	GCA }	GAA } Glu	GGA }
		GUG }	GCG }	GAG }	GGG }

Table 1. Genetic code.

The “Wobble” Hypothesis

Even before the genetic code had been elucidated, Francis Crick postulated that base pairing of the mRNA codons with the tRNA anticodons would require precision in the first two nucleotide positions but not so in the third position (the precise conformation of **base pairs**, which refers to the **hydrogen bonding** between A-T (A-U in RNA) and C-G pairs is known as Watson-Crick base pairing). The third position, in general, would need to be only a purine (A or G) or a pyrimidine (C or U). Crick called this phenomenon “wobble.”

This less-than-precise base pairing would require fewer tRNA species. For example, tRNA^{Glu} could pair with either GAA or GAG codons. In looking at the codon table, one can see that, for the most part, the first two letters are important to specify the particular amino acid. The only exceptions are AUG (Met) and UGG (Trp) which, as indicated above, have only one codon each.

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

hydrogen bonding weak bonding between the H of one molecule or group and a nitrogen or oxygen of another

The Code Has No Gaps or Overlaps

The 1960s were an exciting time for molecular biologists, for it was then that the genetic code was broken. Two possibilities had to be considered for the genetic code. It was possible that the code had gaps, that is, some sort of punctuation mark or a “spacer” nucleotide or nucleotides between coding groups. Second, the code could be either overlapping or nonoverlapping. These possibilities are illustrated in Figures 2 and 3. An overlapping code would have the advantage that more information could be contained in a smaller space.

mRNA sequence: A U G C C U U A G

Protein sequence: Met Pro Leu
Cys Leu Stop
Ala Phe

Figure 2. Any stretch of messenger RNA has three different reading frames, which can be translated to give different amino acid sequences. Only one of them is the correct sequence, dictated by the start point of the first triplet, AUG.

Figure 3. Models of mRNA showing the effect of hypothetical “punctuation marks” separating codons.

Punctuated mRNA:	A U G p C C U p U U A p G U C p . . .
Protein sequence:	Met Pro Leu Val
Punctuated mRNA:	A U G p C G C U p U U A p G U C p . . .
Insertion of a base:	▲
Protein sequence:	Met Arg Leu Val

However, in overlapping code a mutation that changed one base would lead to the changing of three consecutive amino acids in the protein sequence. Genetic evidence, available even before the code had been deciphered, indicated that a single point mutation, that is, a change in a single nucleotide, affected only one amino acid and thus suggested a nonoverlapping code.

Another possibility was that the code had punctuation marks, that is, a base (indicated by “p” in Figure 3) acting as a comma that would separate each codon. In this situation, if an additional base were inserted into a codon, then only that codon would be affected. In a code without punctuations or gaps, however, insertion of a single nucleotide would result in all codons from that point on being affected. This would in turn change the amino acid sequence in the protein from that point on. Again, genetic evidence ruled out a punctuated code, as base insertions do, in fact, affect the entire protein from the insertion point on, rather than just a single amino acid. This effect is called a frameshift mutation.

In the late 1970s DNA sequencing techniques were developed. A number of proteins had already been sequenced by protein sequencing methods. When the genes for these proteins were cloned and sequenced, the predicted protein sequence could be deduced. Agreement between the DNA and protein sequences confirmed the accuracy of the genetic code.

Exceptions to the Universal Genetic Code

After the original genetic code of *E. coli* was completed in 1968, the genetic code was subsequently determined for many other organisms ranging from bacteria to mammals, including humans. The codons were found to be the same for all organisms, leading to the idea that the genetic code is “universal.” Furthermore, it also suggested that life on Earth had a single evolutionary origin, otherwise there would have been numerous genetic codes. The code was established during evolution, probably by chance, as there are no compelling reasons one codon should prevail over another. After it was established, any subsequent changes in the code would prove to be lethal, for if one codon changed, then all similar codons in the entire organism’s genome would have to change simultaneously—a highly unlikely possibility.

Thus, it was surprising to find that there are, in fact, a few rare exceptions to the universal code. These exceptions are listed in Table 2. Most of these exceptions are found in the mitochondrial genome. The mitochondrion is thought to have evolved from an **endosymbiotic** bacterium at the time when the eukaryotic cell first arose. The mitochondrial **genome** is small, and most of the genes of the original endosymbiont have migrated to the nucleus.

E. coli the bacterium
Escherichia coli

endosymbiotic a type of symbiosis in which one partner lives within the other

genome the total genetic material in a cell or organism

EXCEPTIONS TO THE UNIVERSAL GENETIC CODE

Organism	Normal codon	Usual meaning	New meaning
Mammalian mitochondria	AGA, AGG	Arginine	Stop codon
	AUA	Isoleucine	Methionine
	UGA	Stop codon	Tryptophan
<i>Drosophila</i> mitochondria	AGA, AGG	Arginine	Serine
	AUA	Isoleucine	Methionine
	UGA	Stop codon	Tryptophan
Yeast mitochondria	AUA	Isoleucine	Methionine
	UGA	Stop codon	Tryptophan
	CUA, CUC, CUG, CUU	Leucine	Threonine
Higher plant mitochondria	UGA	Stop codon	Tryptophan
	CGG	Arginine	Tryptophan
Protozoan nuclei	UAA, UAG	Stop codons	Glutamine
<i>Mycoplasma capricolum</i> bacteria	UGA	Stop codon	Tryptophan

Table 2.

In examining the exceptions to the universal genetic code in Table 2, you can see that there are only a few changes, most notably the use of a standard “stop” codon to encode an amino acid. For example, UGA normally is a stop codon. But in the mitochondria of the fruit fly *Drosophila melanogaster*, it encodes the amino acid tryptophan.

A few additional exceptions to the universal genetic code have also been identified. These include the nuclear genome of a few protozoan species and also in the bacterium *Mycoplasma capricolum*. These exceptions, however, do not imply multiple evolutionary origins of life. What is most striking is that the “exceptional” meanings of most of the codons are identical across all the organisms in which they are found, not different. Had there been multiple origins, we would expect to see drastically different genetic codes in these exceptional organisms. SEE ALSO CRICK, FRANCIS; *ESCHERICHIA COLI* (*E. COLI* BACTERIUM); NUCLEOTIDE; READING FRAME; RIBOSOME; TRANSCRIPTION; TRANSLATION.

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Genetic Counseling

Over the last half-century, our understanding of genetic disorders has increased spectacularly. When facts about inherited disorders first came to light, health professionals began to inform families about probable inheritance patterns and recurrence risks (the likelihood that offspring or other relatives might also inherit the disease).

A couple receives genetic counseling during their sixth month of pregnancy. Genetic counselors are specially trained to provide support to couples whose families have histories of genetic diseases, or who are pregnant and carrying an affected child.



The Need for Genetic Counseling

Receiving a diagnosis of a genetic disorder can have profound impact for both patients and their family members, and it quickly became clear that aside from the need for medical and genetic information, families affected by genetic disorders had educational, social, and psychological needs that required attention. And though families were afflicted with different disorders passed on by different modes of inheritance (**autosomal** recessive, **autosomal dominant**, complex, or some other type), certain reactions were observed again and again.

For instance, some parents of children with an **autosomal recessive** disorder felt profound guilt at having transmitted an inherited disorder to their

autosomal describes a chromosome other than the X and Y sex-determining chromosomes

dominant controlling the phenotype when one allele is present

recessive requiring the presence of two alleles to control the phenotype

child. On the other hand, a family member who was spared a genetic disorder that other family members developed frequently suffered “survivor guilt.” Health-care providers also noted that family members at risk for developing a late-onset disorder live with intense anxiety about the future and often needed support and counseling. Finally, counseling was seen to be of potential help for family members who incorrectly inferred that they were at risk for having a child with a genetic disorder. Attempts to meet these varied needs and help give people a sense of control over their situation resulted in the emergence of a model of genetic education and support that came to be called genetic counseling.

An Evolving Field

In time, genetic counseling evolved into a profession. Since the early 1970s genetic counselors have been members of health-care teams providing comprehensive and consistent medical genetic services, while also tending to the social and emotional welfare of the patients and their families. In the United States, the first master’s degree training program for genetic counseling was established in 1971. Since then the profession has grown tremendously. There are now more than 2,500 genetic counselors in the United States and 25 genetic counseling training programs. Many industrialized countries have adopted the United States’s model of training for genetic counselors, and master’s-level training programs now exist in Canada, Australia, Great Britain, and South Africa.

As the profession grows, the definition of genetic counseling also continues to evolve. Genetic counseling is currently defined as “a communication process, which helps an individual and/or family in a variety of ways.” For instance, genetic counselors help patients and their families to comprehend the medical facts, including the diagnosis, probable course of the disorder, and available treatment options. Genetic counselors also help educate their clients about the way heredity contributes to the disorder and the risk of recurrence in relatives, and to understand the options available for dealing with this risk of recurrence.

Genetic counselors also teach their clients the medical facts relating to a disorder, enabling them to make informed, independent decisions. They understand that only if their clients possess the necessary facts about available medical care and genetic testing can their decisions be free of coercion. Finally, genetic counselors provide information helpful in accessing local and national support resources.

A key aspect of the genetic counselor’s work is educational: helping clients to comprehend the genetic implications of their disorder. In addition, the diagnosis of a genetic disorder in an individual often leads to identification of other family members who may be at risk for having or passing on a genetic disorder, so genetic counselors often work with entire families. For instance, if the genetic tests of a female patient with two sisters disclose that she has a genetic change (mutation) in the BRCA1 gene, then her two sisters are at risk for carrying this same genetic change, which can cause breast cancer. Once the patient has been notified and has given her permission, these sisters would then be contacted and given the chance to learn about their own risk of carrying a disease-causing gene.



A visual representation of the various services offered by genetic counselors.

SPECTRUM OF GENETIC SERVICES PROVIDED

Genetic education & information



Genetic educators,
Single disorder counselors
(e.g. for sickle cell disease)

Physicians,
Nurses,
Social workers

Genetic counseling

Genetic counselors,
Medical geneticists

This second round of counseling is important. Armed with the information about their susceptibility for breast cancer, the sisters might choose to undergo genetic testing themselves, or they might begin early detection screening evaluations. Their new knowledge might also lead them to adopt lifestyle changes that could reduce their risk of developing breast cancer. Even when genetic testing is not available, early identification of at-risk patients and their family members can be valuable and quite possibly lifesaving.

The demand for genetic education and counseling will likely increase as knowledge accumulates about the genetic component of commonly occurring disorders such as breast cancer, Alzheimer's disease, heart disease, diabetes mellitus, and **osteoporosis**. As a result, a variety of professional specialists, such as genetic educators, physicians, nurses, social workers, medical geneticists, and genetic counselors, will increasingly be called upon to provide genetic education and counseling. Of this group, however, genetic counselors and medical geneticists are the most qualified to perform comprehensive genetic counseling.

Protection as Well as Education

At the same time that they provide beneficial genetic counseling to patients and their families, professionals providing such a service must have a full understanding of the dangers of **eugenics**. The abuse of genetic information has led to many atrocities in the past. In Germany, the Nazis murdered nearly 7 million "genetically defective" people during World War II and forcibly sterilized nearly half a million others, all in the name of "eugenics"—a policy that calls for the systematic elimination of "unfit" members of the population. The United States also has a checkered past with respect to eugenics. In the early twentieth century, the United States passed laws allowing sterilization of the mentally handicapped and limiting the number of "genetically inferior" ethnic groups that were allowed to immigrate.

In order to prevent such abuses from ever occurring again, the genetic counseling profession has followed in the footsteps of other health-care professions by establishing a code of ethics guiding professional behavior. Policies such as nondirectiveness, prevention of genetic discrimination, respect for patients' beliefs, complete disclosure, and **informed consent** are components of these ethical principles. Nondirectiveness, one of the major tenets of genetic counseling, is defined by the National Society of Genetic Counselors as enabling "clients to make informed independent decisions, free of coercion, by providing or illuminating the necessary facts and clarifying the alternatives and anticipated consequences."

osteoporosis thinning of the bone structure

eugenics movement to "improve" the gene pool by selective breeding

informed consent knowledge of the risks involved

Genetic counseling has become a vital part of medical genetics. With the knowledge gained from the past and the tools to help patients choose their paths, genetic counseling will continue to be invaluable in the rapidly growing field of human genetics. SEE ALSO EUGENICS; GENETIC COUNSELOR; GENETIC TESTING; INHERITANCE PATTERNS; PRENATAL DIAGNOSIS.

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Genetic Counselor

Genetic counselors are health professionals trained in genetics, genetic disorders, genetic testing, molecular biology, psychology and psychosocial issues, and the ethical and legal issues of genetic medicine. Most genetic counselors have a master's degree from a genetic counseling training program. The very first class of genetic counselors was graduated from Sarah Lawrence College in 1971. There are about 2,000 genetic counselors in the United States. Most are women under the age of forty, but the field is becoming more diverse.

Genetic counselors are board-certified by the American Board of Genetic Counseling. Board eligibility or certification is required for employment in many positions, and some states are beginning to license genetic counselors. While salaries vary significantly by geographic location, years of experience, and work setting, according to a Professional Status Survey conducted by the National Society of Genetic Counselors, Inc. (NSGC) in 2000 the mean salary for a full-time master's-level genetic counselor was \$46,436. The NSGC, incorporated in 1979, is the only professional society dedicated solely to the field of genetic counseling. Its mission is "to promote the genetic counseling profession as a recognized and integral part of health care delivery, education, research, and public policy."

The role of the genetic counselor has evolved greatly since 1971. Initially, genetic counselors worked almost exclusively in the clinical setting under physician supervision, seeing clients who had been diagnosed as having a genetic disorder, were at risk for developing a genetic disorder, or were at risk for having a child with a genetic disorder. They would assess genetic risk, provide information, discuss available testing options, and provide



patient advocate a person who safeguards patient rights or advances patient interests

appropriate supportive counseling. The variety of patients and the information and testing options offered by genetic counselors was greatly restricted by the limited technology and genetic knowledge of the time.

Today, as a result of the Human Genome Project and other advances, genetic counselors are now able to offer more services and options. They are able to specialize in a particular area of interest, such as cancer, prenatal, pediatric, assisted reproduction, and metabolic or neurogenetic disorders. Most genetic counselors (more than 80 percent) still work in the clinical setting, either in a hospital or in private practice. However, advances in genetics have enabled genetic counselors to work in a variety of other settings including research, public health, education, and industry.

As a **patient advocate**, the genetic counselor also remains informed of ethical and legal issues regarding the use of information generated by the Human Genome Project and incorporates pertinent information into the counseling session. For example, the decision to undergo genetic testing may involve controversial issues. Depending on the type of test and the disorder present, testing may have implications for other family members, insurance eligibility or coverage, employment, and quality of life. It is the role of the genetic counselor to ensure that clients are aware of concerns relevant to their situation.

Opportunities for the genetic counselor also exist to consult on research projects, guest lecture, publish articles and books, and teach. Broad training makes genetic counselors highly adaptable to virtually any setting where genetic information is utilized. Overall, genetic counseling is a dynamic and evolving profession. SEE ALSO GENETIC COUNSELING; GENETIC TESTING; GENOMIC MEDICINE; POPULATION SCREENING.

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Internet Resource

National Society of Genetic Counselors, Inc. <<http://www.nsgc.org>>.

Genetic Discrimination

The potentially stigmatizing nature of genetic information and its history of abuse necessitate special provisions for its protection. Although the U.S. Supreme Court has established a basic right to privacy (in the case of *Roe v. Wade*), the nonspecific nature of this privacy protection fails to adequately guard against the unauthorized disclosure of personal genetic information.

There have been reports that genetic discrimination has been practiced by employers and insurers who are afraid of being required to cover the potentially high medical expenses of individuals who have a family history of a genetically inheritable disease, even if those individuals exhibit no symptoms of the actual disease. However, recent debate suggests that the actual occurrence of genetic discrimination has not yet been proven and that the perception of risk is exaggerated. Nevertheless, researchers working in

human genetics should proceed under the assumption of potential harm and should therefore keep confidential all information that they collect or generate as part of any family study they conduct.

Antidiscrimination Legislation

The Rehabilitation Act of 1973 was the first law to prohibit employment discrimination by federally funded agencies and institutions based on physical disability. The 1992 reauthorization of this legislation defines an “individual with a disability” as any person who “is regarded as having impairment.” Such a definition is broad enough to include an **asymptomatic** gene carrier who is perceived as being “sick” by an employer or insurer. For example, carriers of the trait for sickle cell disease may be “regarded” as having the disease, even if they show no symptoms of it. In some cases, such individuals have been denied health insurance because they were inappropriately viewed as having a preexisting condition, that is, as actually having the disease, rather than simply carrying the gene for it.

The 1990 Americans with Disabilities Act (ADA) significantly broadened the scope of the 1973 legislation by prohibiting discrimination against disabled individuals in most areas of employment and with regard to access to public transportation. While the ADA’s definition of disability excludes people with a “characteristic predisposition to illness or disease,” some professionals believe that the legislation allots sufficient protection against genetic discrimination in the workplace.

Nonetheless, there is disagreement about whether people with a genetic predisposition to disease are adequately protected under the ADA. However, a recent ruling by the Equal Employment Opportunity Commission has interpreted the ADA to include the protection of individuals from employer discrimination based on genetic test results. The debate has not yet been fully resolved. For instance, there is still the potential for employers to institute genetic screening programs prior to offering employment. Thus, the municipal, state, and federal judiciaries will be called on to define the extent of the ADA legislation in their respective precincts in future cases.

Although the ADA prohibits employer discrimination, this law, along with the majority of state laws, does not adequately protect those with genetic disorders, whether symptomatic or asymptomatic, against discrimination by insurers. Thus, even though employers are prohibited from discriminating against workers with genetic disorders, the employment opportunities for such persons may still be limited. This is because employers could refuse to hire such individuals on the grounds that their insurance premiums would be too high.

Several states have attempted to address this problem by passing laws protecting individuals from genetic discrimination by health insurers. However, these laws generally fail to offer comprehensive protection to all people at risk for genetic discrimination. Depending on the way an individual law is worded, healthy gene carriers; people who are predisposed to certain genetic disorders such as cancer; and people who are at a high risk of developing a genetic disorder due to family medical history, rather than because of a detected gene mutation may all be excluded from the protection the law is intended to provide. For instance, Section 514 of the federal Employee

asymptomatic without
symptoms



Retirement Income Security Act (ERISA) exempts self-insured health benefit plans from state insurance laws, so the employees of companies that participate in such plans have no protection. Many employers, especially small businesses, are self-insured, and rely upon commercial insurance companies to administer their health plans. These employer-funded self-insurance plans qualify for the ERISA exemption, and therefore do not have to comply with state mandates for services or state laws regarding genetic discrimination, involuntary testing, or privacy.

Present Problems and Potential Solutions

There is one federal act related to genetic discrimination that overrides ERISA and therefore applies to all health plans, even those offered by small businesses. This federal act is called the Health Insurance Portability and Accountability Act (HIPAA). HIPAA specifies that if employees are covered by a group health insurance policy offered through their employer, they must be offered a similar policy when they change jobs. The new insurance company does not have to offer this coverage at the same rate as was offered by the old plan, but it cannot deny coverage by declaring the genetic disorder to be a “preexisting condition.” In contrast, if a person is either unemployed or self-employed, there is no requirement that an insurer offer him or her a policy.

As information about genetic disorders rapidly expands, the potential harm from genetic discrimination also becomes magnified. As a result, there is increasing recognition of the need for federal legislation guaranteeing a right to genetic privacy. Working groups have been formed to evaluate the impact of genetic information on individual insurability and on the insurance industry. Among these groups are the American Society of Human Genetics’ Ad Hoc Committee on Genetic Testing/Insurance Issues, and the Working Group on Ethical, Legal, and Social Implications of the Human Genome Project, which is jointly sponsored by the National Institutes of Health and the Department of Education. At the end of the twentieth century, congressional committees held hearings on these issues, and in February 2001 a bill was introduced that mandates genetic nondiscrimination in health insurance. SEE ALSO GENETIC COUNSELING; GENETIC TESTING; GENETIC TESTING: ETHICAL ISSUES.

Chantelle Wolpert

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Genetic Drift

Genetic drift is the random change in the genetic composition of a population due to chance events causing unequal participation of individuals in producing succeeding generations. Along with natural selection, genetic drift is a principal force in evolution.

Allele Frequencies

Different forms of a gene are called **alleles**. Individual members of a population have different alleles. Together, all the alleles for all the genes in a population constitute the “gene pool” of the population. Through reproduction, individuals pass their genes on to the next generation. If considering only the effect of genetic drift, the larger the population is, the more stable the frequency of different alleles in the gene pool will be over time. In small populations allele frequencies are likely to change rapidly and dramatically over very few generations, or “drift,” because of chance events. This rapid change can occur in small populations because each individual’s alleles represent a large fraction of the gene pool, and if an individual did not reproduce it could have a much larger effect than in the case of an individual in a large population not reproducing. Also, alleles that are found infrequently are more likely to be lost due to random chance.

After many generations, if only genetic drift is operating, populations (even large populations) will eventually contain only one allele of a particular gene, becoming “monomorphic,” or fixed for this allele.

Random Events

Many types of random events that can affect the likelihood of alleles being passed to future generations can be imagined. An adult may fail to mate during mating season due to unusually adverse weather; a pregnant mother may discover a rich food source and produce unusually strong or numerous offspring; all the offspring of one parent may be consumed by predators. Many other scenarios are possible.

To see how such events affect allele frequencies, imagine a population that contains four individuals of an organism that reproduces once and dies. Let us examine how allele frequencies change for a gene that has two alleles, A and a. As with other genes, each individual has two alleles, one inherited from each parent. Imagine that three of the individuals are aa **genotype**, and one is Aa genotype. Thus, of the population’s eight copies of the gene, one is A, and seven are a. Now imagine that because of random chance, the Aa individual does not reproduce. Therefore, only aa offspring are produced and the A allele is lost to the population. The A allele goes from a frequency of one-eighth to zero through the process of genetic drift.

A large reduction in population size can lead to a situation known as a genetic bottleneck. After a genetic bottleneck the population is likely to have different allele frequencies. When only a very small number of individuals are left after a population decline, the population will have only the alleles present in these few individuals. This is known as the “founder effect.” The founder effect can be viewed as an extreme case of a genetic bottleneck. If a population decline affects all individuals in the population without respect to the alleles they carry, genetic drift will have an effect on all genes.

Genetic drift has important implications for evolution and the process of **speciation**. When a small group of individuals becomes isolated from the majority of individuals of a species, the small group will genetically drift from the rest of the species. Because genetic drift is random and the smaller group will drift more rapidly than the larger group, it is possible that, given

alleles particular forms of genes

genotype set of genes present

speciation the creation of new species



enough time, the small group will become different enough from the large group to become a different species.

The fact that small populations are more subject to genetic drift has important implications for conservation. If the number of individuals of a species becomes small, it becomes increasingly influenced by genetic drift, which may result in the loss of valuable genetic diversity. Conservation biologists seek to maintain populations at sufficient numbers to counteract genetic drift. SEE ALSO CONSERVATION BIOLOGY: GENETIC APPROACHES; FOUNDER EFFECT; GENE FLOW; HARDY-WEINBERG EQUILIBRIUM; POPULATION BOTTLENECK; POPULATION GENETICS.

R. John Nelson

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Genetic Engineering *See Biotechnology*

Genetic Testing

mutations changes in DNA sequences

Genetic testing involves examining a person's DNA in order to find changes or **mutations** that might put an individual, or that individual's children, at risk for a genetic disorder. These changes might be at the chromosomal level, involving extra, missing, or rearranged chromosome material. Or the changes might be extremely small, affecting just one or more of the chemical bases that make up the DNA. In a broader sense, genetic testing includes other types of testing that provide information about a person's genetic makeup, such as enzyme testing to diagnose or identify carriers for a genetic condition such as Tay-Sachs disease.

With hundreds of genetic tests available, determining who should be offered testing and under what circumstances testing should occur is relatively complicated. In general, testing is offered to those at highest risk based on their ethnic background, family history, or symptoms. However, just because genetic testing is possible and a person is at risk, this does not mean it should be offered or will be useful to that person. Genetic testing is unlike other medical tests in that individual results may also provide information about relatives, may be able to predict the likelihood of a future illness for which treatment may or may not be available, may put the person at risk for harm such as discrimination, or may have limited accuracy. There are a number of settings in which genetic testing occurs and within each setting there are a variety of indications and considerations for testing.

Prenatal Genetic Testing

Prenatal (before birth) genetic testing refers to testing the fetus for a potential genetic condition. The pregnant woman is considered the patient and makes decisions regarding prenatal testing. There are a variety of circumstances under which a woman might be offered prenatal genetic testing. The parents of the fetus may have a genetic disorder, or they may be what is known as **carriers**. An abnormality or birth defect may be detected on ultrasound that could indicate a genetic condition. The fetus may be identified to be at increased risk for a chromosome abnormality, such as Down syndrome, or a birth defect, such as spina bifida, based on the result of a maternal serum screening test performed on the mother. This is a test that looks at several proteins made by the fetus that are found in a woman's bloodstream while she is pregnant. Or, the mother might be at increased risk for having a baby with a chromosome abnormality because of her age. While all women are at risk for having a baby with a chromosome abnormality, women who are age thirty-five or older are offered prenatal chromosome testing because the chance their fetus has a chromosome abnormality is equal to or higher than the chance she will have a miscarriage due to the sampling procedure.

As with most genetic testing, prenatal genetic testing should occur in conjunction with genetic counseling. The genetic counselor provides supportive, nondirective counseling and information. Nondirective counseling means that while the counselor will try to facilitate decisions regarding testing and future pregnancy management, she will not make specific recommendations. Because the decision to undergo testing is personal and must take into account differences in beliefs, life circumstances, and the risk of the procedure, the decisions regarding testing and pregnancy management must be made by the patient. This encounter is also likely to include information about risk of the fetus being affected, the disorder in question, and available testing options.

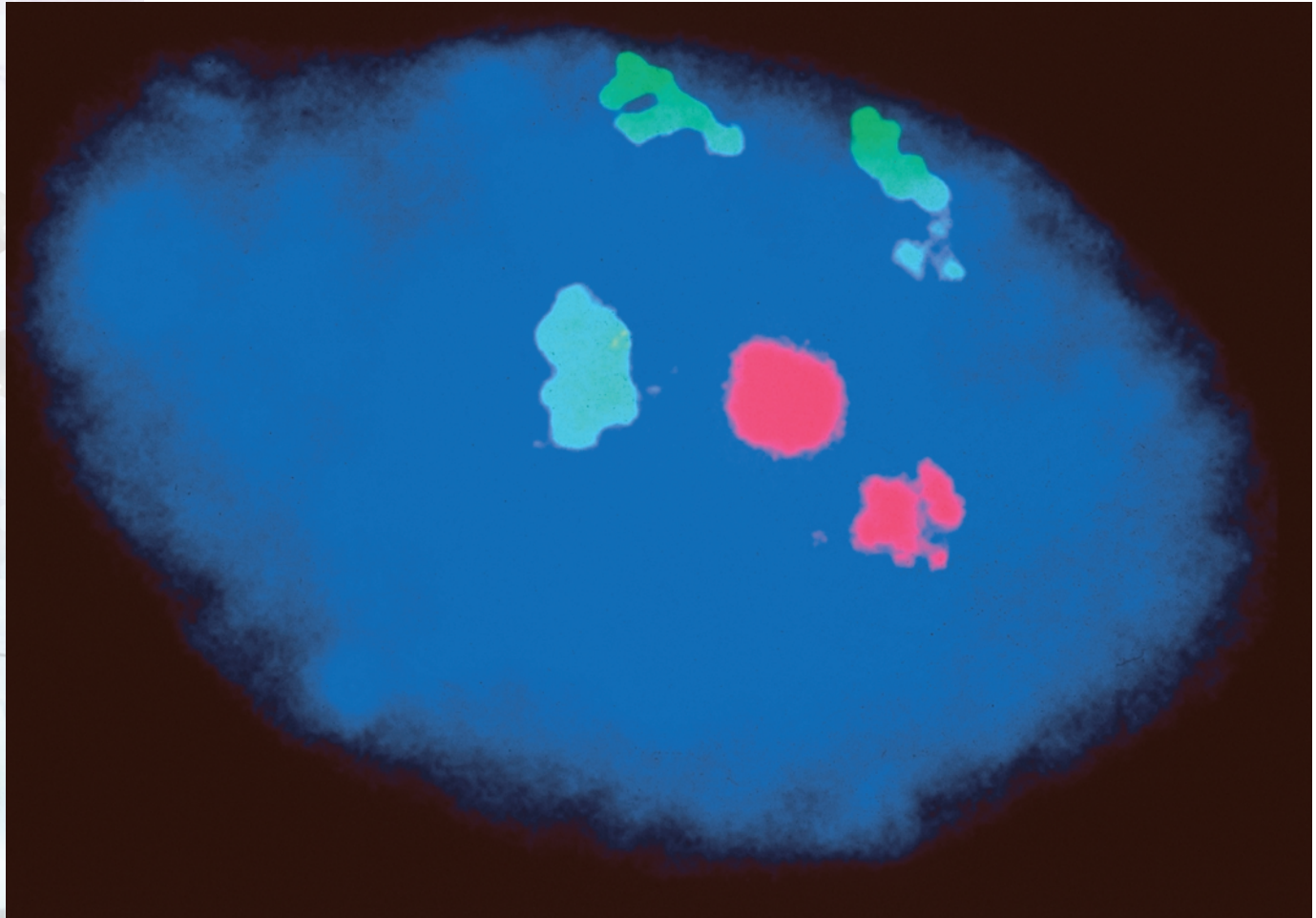
Most genetic tests are performed on tissue or a blood sample. For obvious reasons, obtaining a sample from a fetus is not the same as obtaining one from a child or adult. Prenatal testing procedures are invasive, and there is a risk of miscarriage with every procedure. For this reason, specially trained physicians perform these tests. Prenatal testing can be accomplished using three different methods: amniocentesis, chorionic villus sampling, and **percutaneous** umbilical blood sampling. These tests differ in the type of fetal tissue studied, the timing of the testing during pregnancy, and in their risks and benefits.

Amniocentesis is the most common, and it carries the lowest risk of miscarriage (about one in two hundred pregnancies). It is typically performed between sixteen and eighteen weeks into the pregnancy and involves collecting a small amount of amniotic fluid that contains cells of the developing fetus, which can be used for testing. Chorionic villus sampling is performed earlier than amniocentesis, typically between ten and twelve weeks of pregnancy, but about one in one hundred pregnancies are miscarried as a result of this procedure. It involves obtaining a small sample of chorionic villi (fingerlike projections of the chorion, a membrane that will later develop into the placenta), which should contain cells of the fetus. Percutaneous umbilical blood sampling, typically performed after eighteen

carriers people with one copy of a gene for a recessive trait, who therefore does not express the trait

percutaneous through the skin





This genetic test is positive for Trisomy 18. Fluorescence *in situ* hybridization reveals three copies of chromosome 18 (green) and female sex chromosomes (fuschia) within a nucleus (blue).

in vitro “in glass”; in lab apparatus, rather than within a living organism

weeks, is the most difficult to perform and carries the highest risk of miscarriage (about one in fifty pregnancies). It involves withdrawing blood from the umbilical cord and is primarily used when results are needed extremely quickly, or when only a fetal blood sample can provide a given answer about the fetus. For example, it may be used to test the fetus if the mother has been exposed to an infectious organism known to cause birth defects.

Assisted Reproduction

While **in vitro** fertilization has been available for over two decades, more recently it has become possible to test the resulting embryos for genetic disorders when the embryos are between eight and sixteen cells in size. In this procedure, one to two cells are removed, and the section of DNA containing the gene in question is replicated and tested. Only those embryos identified to be free of risk (based on the DNA results) for developing the genetic disorder are implanted in the uterus. This technique is not widely available, however, and it is both expensive and time consuming. Thus, it is used only infrequently.

Newborn Screening

Newborn screening is unique in being the only genetic testing that it is mandated by the state. The premise of newborn screening is that, for some dis-

orders very early detection and initiation of treatment will prevent health problems, often mental retardation. All newborn infants are tested for a variety of genetic disorders. Each state determines for itself for what disorders to test their newborns. Disorders are chosen based on severity, incidence, ease and accuracy of testing, cost, and benefit of early diagnosis. All states test newborns for phenylketonuria (PKU), a metabolic disorder that is almost never evident at birth. Individuals with PKU are missing an enzyme called phenylalanine hydroxylase, which results in the buildup of phenylalanine. If left untreated, severe mental retardation develops. However, infants with PKU who are placed on a diet low in phenylalanine immediately after birth are expected to develop normally, making PKU an excellent candidate for newborn screening.

Symptomatic Genetic Testing

Genetic testing of individuals who are exhibiting symptoms of a genetic disorder is relatively straightforward. Testing is necessary to either make or confirm a diagnosis, which may improve treatment and establish risk estimates for other family members.

A concern related to symptomatic testing is the duty to recontact the patient in the future if more information becomes available. Often, symptomatic individuals, usually children, present with a variety of symptoms for which no diagnosis can be made clinically and for which there is no genetic test. However, with the completion of the Human Genome Project and the wealth of research being conducted, new genes are discovered regularly, which may result in new testing possibilities. Most physicians inform their patients that more information may be available in the future and ask their patients to contact the clinic periodically to inquire about such updates. It is unclear whether this is sufficient to fulfill the physicians' obligation; however, no clear standards exist on this issue.

Carrier Testing

Another common testing situation is carrier testing for autosomal recessive disorders. Autosomal recessive disorders are caused by the inheritance of two nonfunctioning genes, one from each carrier parent. The parents are referred to as carriers because they carry only one nonfunctioning gene and are, therefore, not affected by the disorder. Every individual is thought to be an unaffected carrier of some autosomal recessive disorder. This is only a problem, however, if two individuals who both carry the same recessive disorder conceive a child together. Under this circumstance, the child would have a one in four (25%) chance of inheriting a nonworking copy from each parent, thereby inheriting the disorder.

There are hundreds of genetic tests available, but it is not practical to perform every available test on each person. Carrier testing is typically offered only to those individuals who are at increased risk based on family history or ethnic background. While family history bears an obvious correlation, ethnic background is important because those who descend from the same group of ancestors are more likely to carry the same genetic changes. For example, individuals of Ashkenazi Jewish descent have about a one in thirty chance of being a carrier for a condition called Tay-Sachs disease,



whereas the carrier frequency is only about one in 300 in other populations. In instances such as this, population screening is often recommended.

Presymptomatic Testing

Presymptomatic testing (that is, testing a healthy person before symptoms appear) may be considered for a genetic disorder for which there is a family history. The decision to undergo this type of testing is not usually straightforward and should always be accompanied by genetic counseling. There are a number of considerations to take into account when deciding whether to proceed with testing. The first is the usefulness of the information. How will knowing the genetic information benefit the person? Testing is more favorable when preventive treatment is available, when results might have a significant impact upon life decisions, such as having children or getting married, or if it will ease extreme anxiety to learn one's genetic status. If no treatment is available, as in the case of Huntington's disease and other triplet repeat diseases, the information may be of less benefit. In some cases it may even be psychologically harmful.

mutations changes in DNA sequences

The second consideration is accuracy, not only of the actual test result, but also of its ability to predict the development of the disorder. Some disorders are caused by more than one gene, or by multiple changes or **mutations** in the same gene. A given test might not be able to look at all mutations or every gene that causes a disorder, leaving a person who tests negative with doubt as to whether they are truly mutation-free. Some genetic tests, particularly those for complex disorders, are for susceptibility genes. As the name implies, these are genes that make a person susceptible to developing a disorder, but do not guarantee it. An example of this is breast cancer. When deciding whether or not to test for such a disorder, it is important to ask how it would feel to test positive for a susceptibility gene for a serious genetic disorder that may never develop.

The third consideration is risk of personal harm. Testing positive for a disorder may put a person at risk of economic or social harm. Although rare, there have been instances where individuals have been denied insurance, employment, or both based on the results of genetic testing. Also, a person may be at risk of experiencing psychological or emotional problems after undergoing genetic testing. There have been instances, particularly with Huntington's disease testing, where individuals have committed suicide following a positive test result. All of these factors must be presented to, discussed with, and weighed by the individual considering testing, in the context of genetic counseling and prior to making a decision about testing.

Presymptomatic Testing of Children

Presymptomatic testing of children has somewhat different considerations. It is typically considered only when the onset of the disorder occurs in childhood, or when knowing the genetic status will significantly benefit the child, for example by enabling him to receive early preventive treatment. For example, children at risk for inheriting the gene that causes retinoblastoma (cancer of the retina) may be tested because the disease usually presents before age five. With early treatment, the long-term outcome is favorable. Know-

ing whether the child has inherited the gene will allow physicians to know whether to aggressively screen the child for signs of cancer development.

Many genetic professional organizations have developed position statements regarding genetic testing of children that discourage testing for disorders that do not pose a risk in childhood and for which early identification poses no benefit to the child. This includes adult-onset disorders, such as Huntington's disease, but also pertains to carrier testing of females for X-linked recessive disorders, such as muscular dystrophy. In most X-linked (sometimes referred to as sex-linked) recessive disorders, females who inherit a mutation on one of their two X chromosomes are usually unaffected carriers because the second X chromosome is able to compensate for the loss. However, because males have only one X chromosome (the other sex chromosome is a Y), they will be affected if they inherit a mutation. In genetic medicine, personal autonomy is a priority. Individuals have the right to make their own decision regarding genetic testing. If a child is tested for an adult-onset disorder or to determine carrier status, their right to make their own decision as an adult has essentially been taken away. SEE ALSO ALZHEIMER'S DISEASE; BREAST CANCER; CHROMOSOMAL ABERRATIONS; CYSTIC FIBROSIS; DOWN SYNDROME; GENETIC COUNSELING; GENETIC COUNSELOR; GENETIC DISCRIMINATION; GENETIC TESTING: ETHICAL ISSUES; INHERITANCE PATTERNS; METABOLIC DISEASES; MUSCULAR DYSTROPHY; POPULATION SCREENING; PRENATAL DIAGNOSIS; TAY-SACHS DISEASE; TRIPLET REPEAT DISEASES.

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Genetic Testing: Ethical Issues

Genetic testing is the name given to a variety of laboratory techniques, all of which ultimately provide information about a person's underlying genetic makeup, also called their genotype. Genetic information is of interest because it can help in the diagnosis of a current health condition or provide insight into future health.

Advances in Genetic Science

Early understanding of genetic disease was first obtained about "single-gene disorders." These disorders include such diseases as sickle cell disease and





mutations changes in
DNA sequences

cystic fibrosis. They occur because of the strong influence of a mutation in a single gene. Using the tremendous amounts of information about the human genome made available by the Human Genome Project, scientists are now dissecting the genetic components of complex disorders, such as diabetes and heart disease, that come about as a result of both genetic and environmental factors. Information about the genetic factors in complex disorders can help to predict the probability of future development of disease.

Understanding the issue of probability is important for the ethical use of genetic testing technology. “Genetic determinism” describes the idea that genes hold the map of an individual’s future health. Although the detection of some **mutations** can diagnose the presence of a disease, many other tests provide only information about risk for disease, not a certainty that disease will develop. For example, genetic testing to detect mutations in the gene for breast cancer (BRCA1) can give information about an individual’s susceptibility to developing breast and ovarian cancer. However, the probability of developing cancer will depend not only on the status of the gene, but also on factors related to family history and environmental exposures.

A Range of Ethical Issues

Ethical issues that arise within the context of genetic testing are similar to those that arise for any personal medical information. For example, there are concerns about protecting the privacy of the patient and the confidentiality of information, whether it is genetic data or any other item in the medical record. Possibilities of stigmatization or discrimination occur because of social perceptions of some diseases, whether the primary cause is genetic, infectious, or environmental. Furthermore, genetic testing requires informed consent; that is, patients need to be educated about the purpose of the testing (diagnosis, prognosis, or susceptibility assessment), the potential test outcomes and what they mean, and what the options are once the results are known.

Nevertheless, the perspective of “genetic exceptionalism” places genetic information in a category of different, special, or greater concern than other medical information. Knowledge of the genetic information of one person can have an influence on his family members because it implies information about their health status, disease risk, or the possibility of passing a condition on to their offspring. However, while concerns arise from genetic testing because of the obvious familial nature of genetic information, similar concerns arise from the simple knowledge of shared family history or environmental exposures, even without detailed genetic information.

Evaluation of the risks and benefits of genetic testing is an important factor in the process of considering the ethics of its use. The physical risks of genetic testing are generally minimal since, in most cases, DNA can be tested from easy-to-access cells: blood, buccal cells (inside of the cheek), and even hair follicles. However, there is a range of nonphysical risks, including loss of privacy and discrimination.

Consideration of the benefits of a genetic test is sometimes difficult. For example, when a therapy is available for a genetic disease, the availability of this benefit favors conducting the genetic test. If a therapeutic benefit is not available, however, the value of having this information becomes more questionable. Doctors and patients must ask if the infor-

mation will help in some way, such as in life or reproductive planning. Although it can be difficult to compare the value of the information to the possible risks associated with conducting the genetic test, it should be carefully and explicitly considered.

Prenatal and Childhood Testing

Testing in children raises a different set of risks and issues. Adults are able to choose for themselves about genetic testing after hearing the relative advantages and risks. Children, however, must rely on their parents to decide for them. On the one hand, deciding to test children to diagnose a disease seems appropriate in most situations, particularly when a treatment can be provided. On the other hand, there is disagreement about testing children for disorders with adult onset and for which there is no intervention. In this situation, respect for the child's own decision (autonomy) and preserving privacy are ethical arguments for waiting to offer the genetic test. Upon reaching adulthood, individuals may then choose whether to learn their own genetic information.

Genetic testing requires additional considerations when done in the prenatal setting. In this case, parents are deciding to obtain genetic information about a third party, the developing fetus. In cases where there is no available medical therapy for a diagnosed genetic condition, the availability of limited options results in tough ethical choices for parents. The options may range from continuing the pregnancy, preparing for a child with special medical needs, or terminating the pregnancy. To make these decisions, prospective parents need clear information regarding the meaning of test results. Issues to consider include the predictive value of the test, the likely severity and age of onset of the predicted disorder, and the probability that the genetic alteration detected will actually result in a disorder.

Presymptomatic Testing: Huntington's Disease

Presymptomatic genetic testing when no treatment is available stands in contrast to genetic testing done for the diagnosis of a current disorder. Huntington's disease (HD) is an example of a genetic disorder in which clinical manifestations begin in adulthood. Although some symptoms can be managed with medications, no treatment is yet available to alter the gradual loss of muscle control, psychiatric changes, and progressive **dementia**.

HD is an autosomal dominant disorder, which means that an individual with only one abnormal copy of the HD gene will develop the disease, and the children of affected individuals have a 50 percent chance of inheriting the genetic mutation. The gene abnormality occurs when a **polymorphic** CAG repeat sequence is expanded beyond the normal number of 10 to 29 copies. Diagnosis of HD is almost 100 percent sure when the number of CAG repeats is in the range of 36 to 121.

Individuals at risk for HD may want to undergo genetic testing to end the uncertainty of not knowing whether they will be affected. This knowledge may allow for career and life planning. However, concerns that presymptomatic diagnosis of HD would lead to serious psychological distress or suicide led to the development of testing guidelines. The standard was established that all individuals who sought HD testing should receive pretest counseling to explain the test, assess their psychological status, and

dementia neurological illness characterized by impaired thought or awareness

polymorphic occurring in several forms

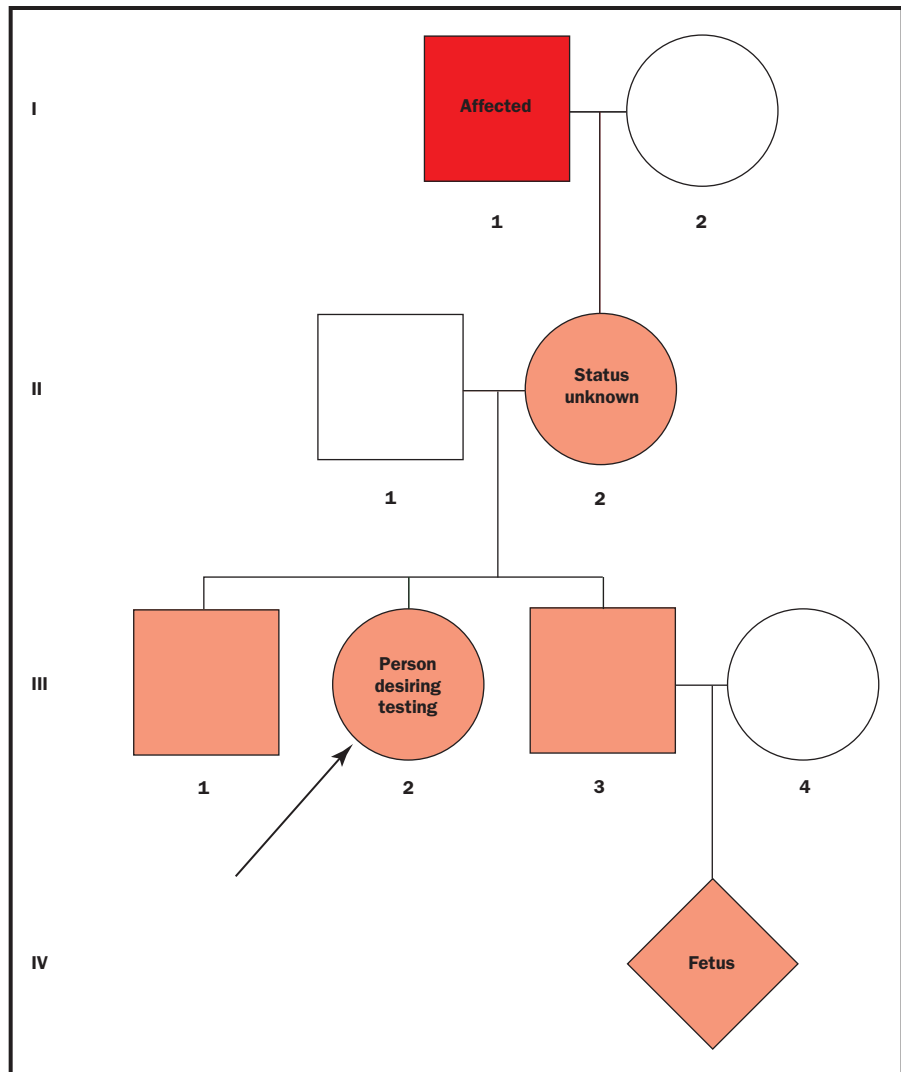


Figure 1. The person desiring the test, known as the proband (arrow; III,2), knows that her grandfather (solid colored square; I,1) had Huntington's Disease (HD). Her mother (II,2) is therefore at fifty percent risk for HD (shaded). If the proband desires to know whether she will develop HD, a direct genetic test can be performed. If the proband is tested and her HD gene is abnormal, then automatically she knows her mother's diagnosis as well, even if the mother does not want to know this information about herself. Similarly, if the proband's sister-in-law (III,4) desires to know the HD status of her fetus (IV), and the child tests positive, a diagnosis of HD could be obtained about her husband (III,3) as well as his mother (II,2) regardless of their preferences for knowing or not knowing their status.

prepare them to consider their possible reactions to the results, whether positive or negative. In addition, results should be given in person, and counseling and support should be available following the test.

Before the identification of a specific gene for HD, genetic testing was done by linkage studies within a family. Linkage analysis required the voluntary participation of affected and unaffected family members across several generations. Identification of the specific gene defect now allows direct genetic analysis, without examining the DNA of any other family member. However, because of the inheritance patterns of HD, information about one family member can give diagnostic information about another. Figure 1 diagrams the ethical complexities arising from situations in which one family

member's genetic test results in necessary implications for other family members. Note that it is not clear whether the individual desiring testing has decisive rights, or if those rights belong to those family members who do not desire such testing.

Susceptibility Testing: Apolipoprotein E

Susceptibility testing is another area that raises possible ethical issues for patients and clinicians. Recall that susceptibility implies that the genetic test does not provide the final answer regarding disease, but rather that additional factors are involved. For example, cardiovascular disease is a major source of **morbidity** and mortality. Polymorphic variants in the apolipoprotein E gene (APOE) have been found to be associated with variations in blood levels of lipids, lipoproteins, and apolipoproteins, which may in turn influence risk for heart disease. Clinicians and patients may wish to learn which APOE variants they have, to help determine their susceptibility to cardiovascular disease and the likelihood that altering their lifestyle risk factors or using pharmaceutical therapy might help prevent cardiovascular disease.

When deciding to test for APOE gene variants, however, there are additional consequences that must be considered, particularly the fact that the gene may play multiple roles in health and disease (**pleiotropy**). Studies have shown that one version of APOE, the e4 **allele**, is associated with an increased risk of Alzheimer's disease. When deciding to undertake genetic testing for the intended immediate medical benefit of understanding cardiovascular disease risk, the individual should therefore also consider the possibility of learning potentially unwanted additional information.

Support Mechanisms

Several support mechanisms exist for dealing with the complex decisions and ethical issues arising from the medical application of genetic testing. In the past, much genetic testing has been carried out by medical specialists with disease-specific expertise who have an understanding of the medical, psychosocial, and family implications of diagnoses of particular genetic diseases. Genetic support groups have also been established for education and mutual support among patients and families with genetic disorders. An organization called the Genetic Alliance provides a referral source to connect individuals with appropriate support groups and information about genetic disorders.

Typically, the best resource for an integrated approach to support for patients and families is a genetic counselor. Genetic counselors provide factual education about genetic disease inheritance, how to understand risk and medical probabilities, and the range of options for care. They also provide counseling to help patients and families to clarify and articulate their own values and motivations in order to make the most appropriate decisions. These may be decisions about whether to undergo a genetic test or how they may act on receiving the results of a test. SEE ALSO BREAST CANCER; GENETIC COUNSELING; GENETIC COUNSELOR; GENETIC DISCRIMINATION; GENETIC TESTING; HUMAN GENOME PROJECT; INHERITANCE PATTERNS; PLEIOTROPY; PRENATAL DIAGNOSIS.

morbidity incidence of disease

pleiotropy genetic phenomenon in which alteration of one gene leads to many phenotypic effects

allele a particular form of a gene

Carol L. Freund and Jeremy Sugarman

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Genetically Modified Foods

A genetically modified (GM) food is a plant that has a genetic change in each of its cells that a researcher has introduced. The modification may add a gene from a different species and thereby create a transgenic plant, or it may overexpress or silence a preexisting plant gene. Overexpression is accomplished by altering the **promoter** region of a gene, which controls how rapidly and in which cells the encoded protein is synthesized, thus directing a plant to manufacture more of a natural product. Conversely, a gene may be "silenced" (directed not to synthesize a protein) through the use of antisense technology, which applies a complementary nucleic acid to messenger RNA, halting expression of the encoded protein.

Genetic Modification in Animals and Plants

Animals have not yet been genetically modified to provide foods. Transgenic animals can, however, produce certain pharmaceuticals, but this approach is still experimental. One possible future use of transgenic animals is to create herds of cattle or sheep that are genetically resistant to developing transmissible spongiform encephalopathies, such as scrapie in sheep and "mad cow disease" in cattle.

Genetic modification in plants produces the same types of changes that result from traditional agricultural techniques, such as controlled breeding. However, genetic modification alters one gene at a time in a controlled manner, and typically has faster results than breeding plants with particular combinations of traits. With standard breeding techniques, it may take a generation to introduce, or remove, a single gene. Breeding a polygenic trait (a trait that involves more than one gene) into apples, which have a generation time of four years, could take two decades or longer.

GM traits that have already been introduced into plants include resistance to insects, insecticides, and herbicides; larger fruits; salt tolerance; slowed ripening; additional nutrients; easier processing; insecticide production; and the ability to take its own nitrogen from the air, lowering reliance on fertilizer. Specific products of genetic manipulation include insect-resistant corn, frost-resistant strawberries, rice that makes beta-carotene (a

promoter DNA sequence to which RNA polymerase binds to begin transcription